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A Further Applications

A.1 Self-interstitials in silicon

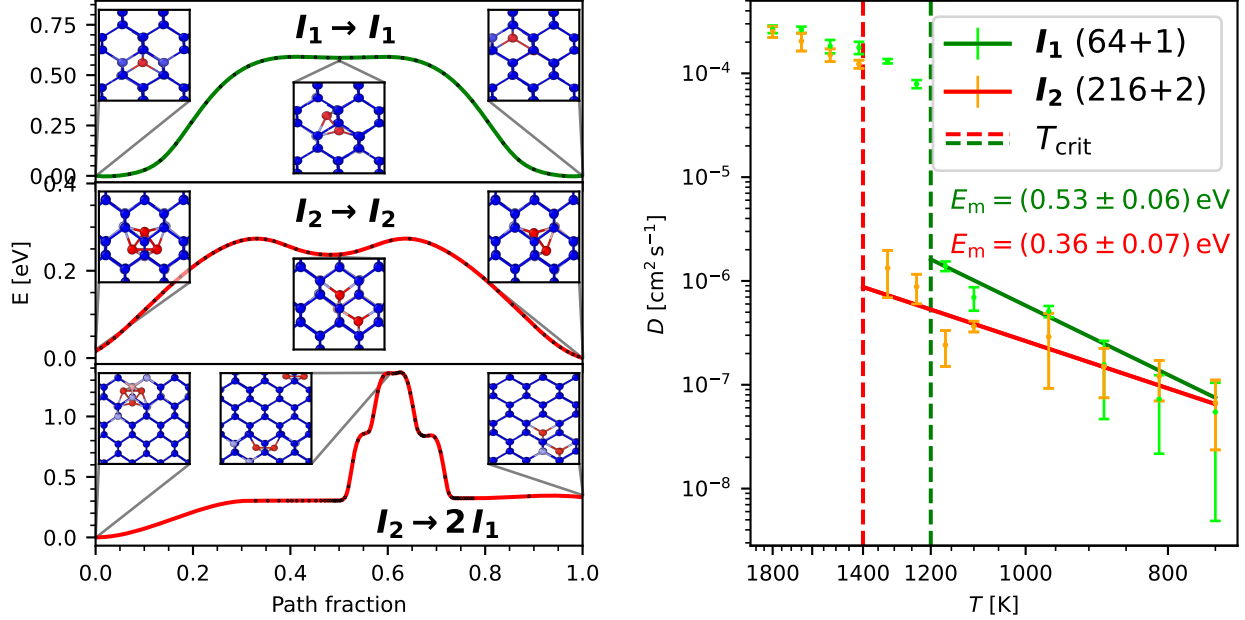


Figure S1: Single- (I_1) and di-interstitial (I_2) defects in silicon. Left: Nudged elastic band paths between two metastable sites with a subset of images shown (black dots). Right: Diffusion coefficient against inverse temperature and Arrhenius laws with migration energy E_m and phonon instability temperature T_{crit} .

Di-interstitial silicon (I_2) constitutes a test case for the transferability of MACE-MP-0 to point defects in a periodic lattice. In the following, self-diffusion coefficients D and interstitial migration energies E_m are obtained from MD simulations without a D3 dispersion correction. Consistency tests with a (64 + 2)-atom silicon structure from (154) relaxed with the PW91 functional are performed. Relaxing with MACE-MP-0 showed no change in energy at a force tolerance of 0.05 eV/Å.

The I_2 structure was generated by relaxing a (216 + 2)-atom diamond structure with lattice constant 5.4 Å. After running NVT MD, the distance to the closest lattice site for each atom was plotted. The interstitials propagate consistent with the accepted mechanism (155). Isolating the trajectories of the point defect shows a characteristic jump length corresponding to expected jumps between stable sites.

At temperatures below 700 K, the interstitials almost exclusively remain bound in a state corresponding to the ground state with C_{1h} symmetry (156) and transition between symmetry-equivalent ground states. At higher temperatures, higher-energy states are sampled in which the interstitials separate into tetrahedral interstitial states, lying at a higher energy 0.3 eV (Fig. S1). We calculate D from the mean-square-displacement (MSD) of all atoms in the unit cell (157). The fit to the Arrhenius law (Fig. S1) gives $E_m = (0.36 \pm 0.07)$ eV, in agreement with (158) and (159).

The above calculations were repeated for a single interstitial (I_1) in a 64-atom cell. The interstitial predominantly occupies the tetrahedral state and transitions between symmetry-equivalent states via the split $\langle 110 \rangle$ state. While these states are expected, the occupancy of each state shows larger deviations from those reported in (157). Figure S1 shows a nudged elastic band (NEB) between two tetrahedral interstitial sites via a $\langle 110 \rangle$ split interstitial site, from which an energy barrier of 0.6 eV is calculated. Repeating the MD simulations at several temperatures (Fig. S1) gives $E_m = (0.53 \pm 0.06)$ eV, in agreement with LDA (157). While the prefactor is strongly system-dependent (160), the migration energy is an intrinsic property of the energy landscape and may be compared with other calculations. However, the finite system size and the interaction between interstitials results in the difference between the single- and di-interstitial migration energies.

MACE-MP-0b3 displays the correct qualitative dependence of the phonon instability temperature T_{crit} on interstitial density (fig. S1). Specifically, $T_{\text{crit}} = 1400$ K of the $(216 + 2)$ -system, which has an interstitial density of approximately 0.9%, is lower than the melting temperature of (1449 ± 10) K calculated for cubic diamond silicon with PBE (161). A further decrease to $T_{\text{crit}} = 1200$ K is observed for the $(64 + 1)$ -system, which has a larger interstitial density of 1.5%. However, the phonon instability temperature for the $(64 + 1)$ -system, calculated with LDA, is 1473 K (157), which MACE-MP-0b3 consequently underestimates. The melting points and phonon instability temperatures are expected to have a dependence on the exchange-correlation functional.

Similarity statement

The MP dataset contains 41 pure silicon structures, including the diamond structure but no self-interstitial defects.

Performance summary

Silicon interstitials display the correct set of local minima, while their relative occupancy at finite temperature is not correct. The predicted activation energies for self-diffusion of the single and di-interstitials are consistent with previous force field and DFT calculations. The dependence on interstitial density is at least qualitatively correct, whereas the phonon instability temperature of the single interstitial system is too low by about 20 %.

Fine-tuning

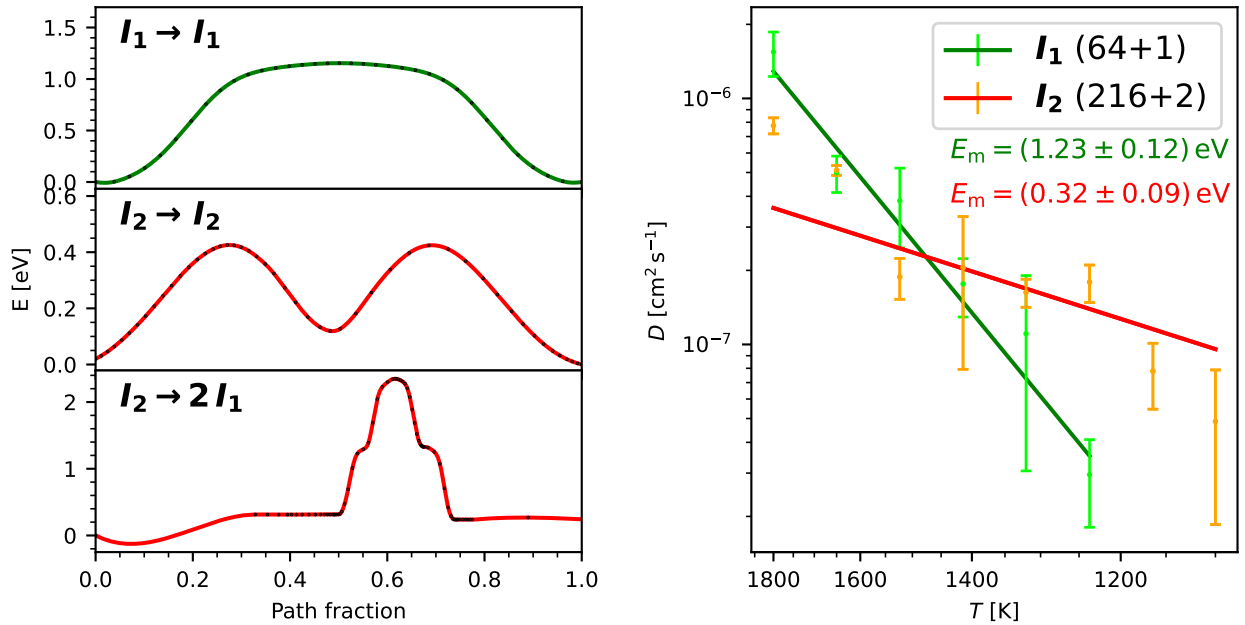


Figure S2: Results for the finetuned model. Plots are the same as in Fig. S1.

Fine-tuning was performed on 100 configurations sampled from an MD trajectory at 1800K from the I_2 system. Fig. S2 shows the NEBs and diffusion coefficients, recomputed with the finetuned model. The barrier heights of all NEBs have increased. The migration energy for I_1 has increased to $E_m = (1.23 \pm 0.12)$ eV, in agreement with the increase in barrier height. However, this is significantly larger than the LDA value in (157). For I_2 , the migration energy has not changed, within the error, and remains consistent with the barrier height. Finally, we note that the phonon instability temperatures for I_1 and I_2 are now above the melting temperature of cubic diamond silicon calculated with PBE (161).

A.2 Amorphous silicon from melt-quench simulations

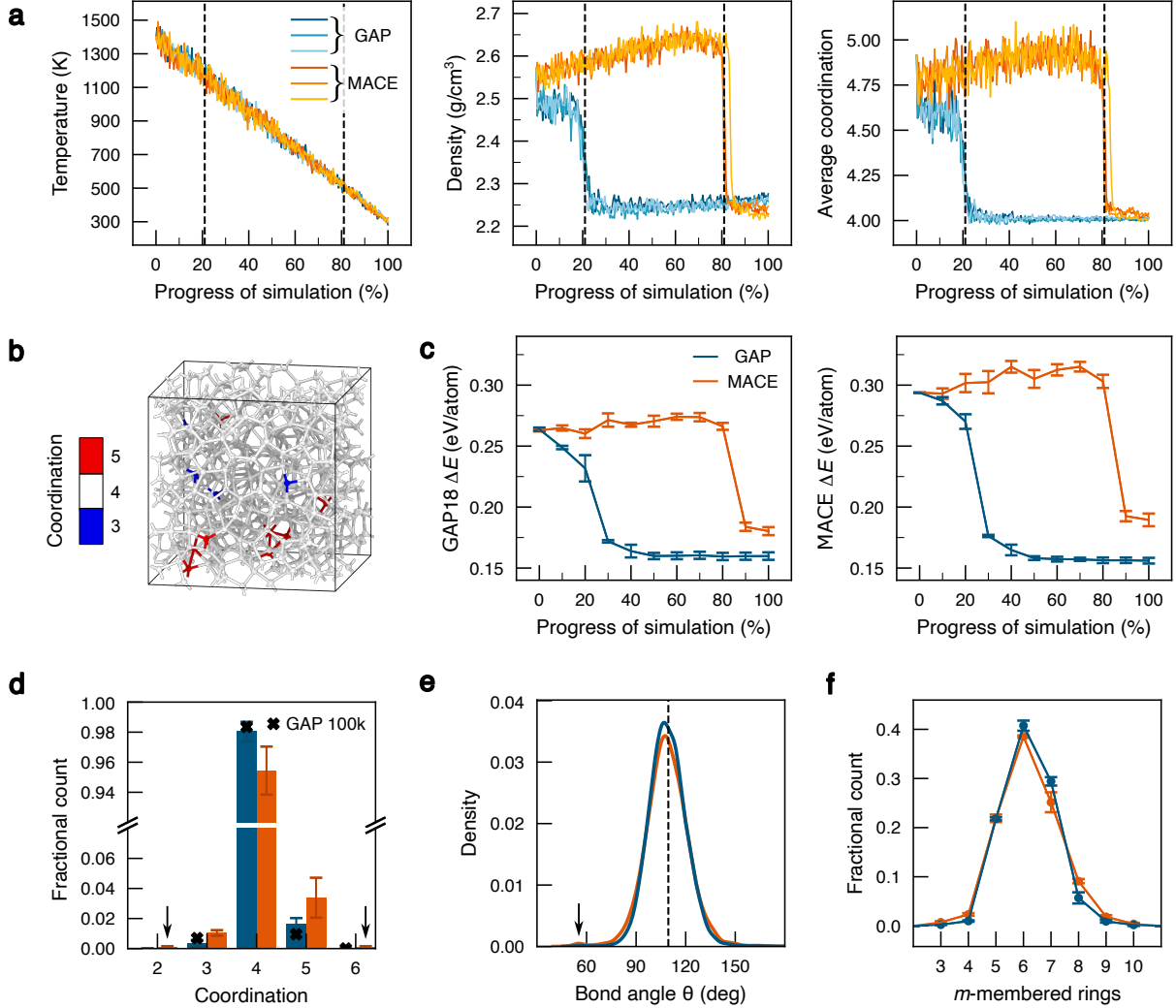


Figure S3: **Amorphous silicon.** We characterise structural models of amorphous silicon simulated by melt-quenching using MACE-MP-0b3 and benchmark against the well-established Si-GAP-18 model (154). (a) Evolution of selected properties during the quench simulations: temperature (left), density (center), and average coordination number, defined using a bond-length cutoff of 2.85 Å (right). (b) Amorphous silicon structure generated using the MACE-MP-0b3 potential, where atoms are color-coded by coordination number. (c) Excess enthalpies (ΔE) relative to diamond-type silicon, computed using Si-GAP-18 (left) and MACE-MP-0b3 (right), for relaxed snapshots from simulations with either potential. (d) Coordination count, where the black crosses represent a study of a 100,000-atom system driven by Si-GAP-18 (162). (e) Bond-angle distribution, where the dashed line is the equilibrium 109.5° angle for tetrahedral environments. (f) Ring-size distribution.

Amorphous silicon (a-Si) is a prototypical disordered material and has served as an example of both the physical and chemical insight afforded by machine-learned potentials (162, 163) and as a challenging benchmark for the development of new methodology and potential models (21, 154, 164).

Here, we assess the performance of the MACE-MP-0b3 potential on standard melt-quench simulations, and compare it to the well-established Si-GAP-18 potential, a general purpose GAP model for Si (154), that has been extensively validated in previous literature (162, 163, 165). We perform melt-quench simulations of

512-atom systems quenching from the liquid state at 1400 K to the amorphous state at 300 K at a rate of 1×10^{12} K/s, for a total of 1100 ps. The simulations were run in LAMMPS (166), in the NpT ensemble using a Nosé-Hoover thermostat and barostat with a time step of 1 fs. A well-equilibrated liquid starting structure was generated by annealing a random hard-sphere model at 1400 K for 10 ps using Si-GAP-18. The protocol was repeated three times for each model to assess the variability in predictions.

Fig. S3 (a) presents the temperature, density and average coordination within a cutoff of 2.85 Å as a function of the progress of the quench simulation. The density of the liquid structure stays constant for the Si-GAP-18 model, but increases slowly to around 2.65 g/cm³ for the MACE-MP-0b3 model, above the experimental density of liquid Si at 2.57 g/cm³ (167). When quenched from the liquid state, Si undergoes a vitrification transition that is accompanied by a sudden decrease in density and concomitantly by a decrease in the average atomic coordination number, as the highly-coordinated metallic liquid transforms into the mainly four-fold coordinated semiconducting a-Si. The MACE-MP-0b3 model predicts a glass transition temperature around 600 K, compared to around 1050 K for Si-GAP-18, and experimental observations of around 1000 K (168, 169). Hence the MACE-MP-0b3 model appears to overstabilise the liquid phase below the experimental glass transition temperature. After the transition, the resulting bulk structures are mostly fourfold-connected, with average densities of 2.266 g/cm³ for Si-GAP-18 and 2.233 g/cm³ for MACE-MP-0b3, close to the experimental density of 2.285 g/cm³ (170).

The final structure from one of the three quench simulations driven by MACE-MP-0b3 is presented in Fig. S3 (b), where the Si atoms are color-coded by coordination numbers. This confirms the insight from Fig. S3 (a), which is that the majority of atoms are 4-fold coordinated, with some 3- and 5-fold coordination defects.

We further compare the excess energies (ΔE), calculated relative to crystalline diamond-type silicon (**dia**), as shown in Fig. S3 (c). These values were computed by taking structural snapshots throughout the quench and relaxing them either with Si-GAP-18 (left) or MACE-MP-0b3 (right), similar to Ref. (165). Both Si-GAP-18 and MACE-MP-0b3 show small energy fluctuations between individual runs for the Si-GAP-18 simulations, while MACE-MP-0b3 had some minor variability in each run. MACE-MP-0b3 predicts the same overall trend in energies as Si-GAP-18, but the energy predictions have an offset of around +0.05 eV/at for the liquid structures, while the predictions from both models on the amorphous structures are very close. The final structures quenched by MACE-MP-0b3 have a higher energy (0.18 eV/at) than those quenched by Si-GAP-18 (0.16 eV/at). This compares to a previous result of 0.14 eV/at for a 4,096-atom system quenched at a rate of 1.0×10^{11} K/s driven by Si-GAP-18 (165) and the experimental excess enthalpy of a-Si after deposition and annealing of 0.14 eV/at (171). The present quench rate is one order of magnitude faster, hence higher average enthalpies and variability can be expected. Nonetheless, the excess enthalpies predicted by MACE-MP-0b3 are qualitatively in the correct ballpark, viz. slightly above the corresponding crystalline phase.

Probing the short range order of the structure, Fig. S3 (d) presents the distribution of Si coordination, averaged over all three repeats for each model. While the Si-GAP-18 structures present fewer than 2% of 3- and 5- fold defects, the MACE-MP-0 structures features around 4% of coordination defects, mostly 5-fold coordinated atoms, but also 2-, 3- and 6-fold defects in smaller concentrations. The 100k-atom simulations in Ref. (162) predicted well-relaxed a-Si to contain on the order of 1.5–2% of defects, consistent with the 4,096-atom simulations of Ref. (165). Faster quenching leads to higher defect counts, as seen from the Si-GAP-18 results in Fig. S3, whereas the overall defect count predicted by MACE-MP-0 is still notably higher than that of Si-GAP-18 at the same quench rate. Another measure of the quality of the short-range structure is the bond angle distribution, shown in Fig. S3 (e), where both distributions appear very similar, but for a small feature at 60° for the MACE-MP-0 structures [arrow in Fig. S3 (e)].

Finally, we assess the quality of the medium-range order of the network using shortest-path ring statistics with the **matscipy** package (172), depicted in Fig. S3 (f). The distribution of m -membered rings is very similar for both models, but the MACE-MP-0 structures have an increased count of small ($m < 5$) and large ($m > 7$) rings, with the presence of $m = 3$ rings that supports the feature in Fig. S3 (e) at 60° and the cluster of 5-fold coordinated atoms in Fig. S3 (b).